CTR 8500-8300

Internet Protocol

(IP) Configuration

Version: 3.16.0

260-668455-001

GA Release

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Americas Technical Help Desk	EMEA Technical Help Desk	Asia Pacific Technical Help Desk
Aviat Networks, Inc. San Antonio, TX U.S.A.	Aviat Networks Motnica 9, 1236 Trzin Slovenia	Aviat Networks Clark Freeport Zone Philippines 2023
Phone: +1 210 526 6345 **Toll Free (USA):** +1 800 227 8332 **Fax:** +1 210 526 6315	**Phone:** +1 210 526 6345 **Fax:** +44 16 9871 7204 (English) +33 1 5552 8012 (French)	**Phone:** +1 210 526 6345 **Fax:** +63 45 599 5196
Email: **TACAM@aviatnet.com**	Email: **TACsupport@aviatnet.com**	Email: **TACsupport@aviatnet.com**

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Conventions and Terminology

This document uses the following conventions and terminology.

Graphic Cues

The following items have graphic cues to identify important supporting information.

Note: A **note** item identifies additional information about a procedure or function.

Caution: A **caution** item identifies important information pertaining to actions that may cause damage to equipment, loss of data, or corruption of files.

Warning: A **warning** item identifies a serious physical danger or major possible problem.

Important: An **important** item identifies additional information that must be heeded.

Font Changes

Bold font style is used for the names of on-screen elements such as; fields, buttons, and drop-down selection lists, keywords, commands and for keys on the keyboard.

Courier font in **blue text** is used to indicate commands that the user needs to type in, with the command prompt shown as brown text.

```
AOS# show radio-carrier status Carrier1/1
```

Any responses or report output from a command is shown as brown text and indented.

```
radio-carrier status Carrier1/1
oper-status up
```

Red text highlights an action, which could be, for example:

- following a menu path
- selecting an option
- clicking a button
- clicking an icon
- pressing a keyboard key.

Common Terminology

Click or **Select:** Point the mouse pointer at the item you want to select, then quickly press, and release the left mouse button.

Right-Click: Point the mouse pointer at the item you want to select, then quickly press and release the right mouse button.

Product Compliance Notes

CTR 8540, CTR 8312, CTR 8311 and CTR 8380 has been tested for and meet all applicable EMC standards. The equipment was tested using screened cable; if any other type of cable is used, it may violate EMC compliance.

CTR 8540, CTR 8312, CTR 8311 and CTR 8380 are Class A products. In a domestic environment this product may cause radio interference in which case the user may be required to take adequate measures. This equipment is intended to be used exclusively in telecommunications centers.

Warning: ODU 600 part number EEH-26-1008-022 must not be co-located with mobile base stations operating in the 800MHz frequency range.

FCC SDoC Statement

CTR 8300/8540 with ODU 600/ODU 600v2/IRU 600v3/IRU 600v4 meets the relevant specifications within FCC CFR 47, Part 101 and FCC CFR 47, Parts 15.107 and 15.109, where applicable. The responsible party, based in the United States, is:

Aviat Networks, Inc.
200 Parker Drive, Suite C100A
Austin, Texas, 78728
Tel: +1 512 265-3680

Regulatory Information for 5.8 GHz band

ODU 600

The following regulatory information applies to license-free operation on the 5.8 GHz band of ODU 600.

FCC Notices

- ODU 600, 5.8GHz, must be professionally installed and maintained.
- This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential environment is likely to cause harmful interference in which case the user will be required to correct the interference at their own expense.
- ODU 600, 5.8GHz, is compliant with the relevant parts of FCC CFR47, Part 15.407.
- To ensure compliance with the FCC RF exposure requirements, a minimum distance of 18 meters must be maintained between the antenna and any persons whilst the unit is operational. This calculation is based on the maximum conducted power and maximum antenna gain.
- ODU 600, 5.8GHz, has been certified for use with a parabolic antenna with a maximum gain of 45.9dBi or a flat panel antenna with a maximum gain of 28dBi.
- The filters and software provided with this product allow for transmission only in the frequency range 5725 – 5850 MHz to ensure compliance with Part 15.407.
- According to the conducted power limit in FCC CFR 47, Part 15.407, the power for this device has been limited to 1W (30dBm) at the antenna port.
- FCC CFR47, Part 15.407 excludes the use of point-to-multipoint systems, omnidirectional applications and multiple co-located intentional radiators. This system is only for fixed, point-to-point operation.

International Use of 5.8 GHz

ODU 600 with CTR 8540, CTR 8312, CTR 8311, or CTR 8380, 5.8 GHz, does not employ DFS, and as such the equipment cannot be deployed within Europe or any country where DFS is a regulatory requirement for protection of radars.

Networking Devices in Electric Power Substations

For IEEE 1613 compliant products, category 7 Ethernet cables must be used in order to ensure compliance.

WEEE Directive

In accordance with the WEEE Directive (2012/19/EU), CTR 8540, CTR 8312, CTR 8311 and CTR 8380 are marked with the following symbol:

This symbol indicates that this equipment must be collected separately for the purposes of recovery and/or recycling.

For information about collection and recycling of equipment please contact your local Aviat Networks sales office. If you purchased your product via a distributor, please contact the distributor for information regarding collection and recovery/recycling.

More information on the WEEE Directive is available at our website:

http://www.aviatnetworks.com/products/compliance/weee/ .

(WEEE is the acronym for Waste Electrical and Electronic Equipment)

RoHS Directive

CTR 8540, CTR 8312, CTR 8311 and CTR 8380 meet the requirements of RoHS directive 2011/65/EU.

Declaration of Conformity, Radio Equipment Directive (RED), 2014/53/EU

Bulgarian	С настоящото Aviat Networks декларира, че този тип радиосъоръжение ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 е в съответствие с Директива 2014/53/ЕС.
Croatian	Aviat Networks ovime izjavljuje da je radijska oprema tipa ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 u skladu s Direktivom 2014/53/EU.
Czech	Tímto Aviat Networks prohlašuje, že typ rádiového zařízení ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 je v souladu se směrnicí 2014/53/EU.
Danish	Hermed erklærer Aviat Networks, at radioudstyrstypen ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 er i overensstemmelse med direktiv 2014/53/EU.
Dutch	Hierbij verklaar ik, Aviat Networks, dat het type radioapparatuur ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 conform is met Richtlijn 2014/53/EU.
English	Hereby, Aviat Networks declares that the radio equipment type ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 is in compliance with Directive 2014/53/EU.
Estonian	Käesolevaga deklareerib Aviat Networks, et käesolev raadioseadme tüüp ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 vastab direktiivi 2014/53/EL.
Finnish	Aviat Networks vakuuttaa, että radiolaitetyyppi ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 on direktiivin 2014/53/EU mukainen.

French	Le soussigné, Aviat Networks, déclare que l'équipement radioélectrique du type ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 est conforme à la directive 2014/53/UE.
German	Hiermit erklärt Aviat Networks, dass der Funkanlagentyp ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 der Richtlinie 2014/53/EU entspricht.
Greek	Με την παρούσα ο/η Aviat Networks, δηλώνει ότι ο ραδιοεξοπλισμός ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 πληροί την οδηγία 2014/53/ΕΕ.
Hungarian	Aviat Networks igazolja, hogy a ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 típusú rádióberendezés megfelel a 2014/53/EU irányelvnek.
Icelandic	Hér með lýsir Aviat Networks yfir því að ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 er í samræmi við grunnkröfur og aðrar kröfur, sem gerðar eru í tilskipun 2014/53/EU.
Italian	Il fabbricante, Aviat Networks, dichiara che il tipo di apparecchiatura radio ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 è conforme alla direttiva 2014/53/UE.
Latvian	Ar šo Aviat Networks deklarē, ka radioiekārta ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 atbilst Direktīvai 2014/53/ES.
Lithuanian	Aš, Aviat Networks, patvirtinu, kad radijo įrenginių tipas ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 atitinka Direktyvą 2014/53/ES.
Maltese	B'dan, Aviat Networks, niddikjara li dan it-tip ta' tagħmir tar-radju ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 huwa konformi mad-Direttiva 2014/53/UE.
Norwegian	Aviat Networks erklærer herved at utstyret ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 er i samsvar med de grunnleggende krav og øvrige relevante krav i direktiv 2014/53/EU.
Polish	Aviat Networks niniejszym oświaDCza, że typ urządzenia radiowego ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 jest zgodny z dyrektywą 2014/53/UE.
Portuguese	O(a) abaixo assinado(a) Aviat Networks declara que o presente tipo de equipamento de rádio ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 está em conformidade com a Diretiva 2014/53/UE.
Romanian	Prin prezenta, Aviat Networks declară că tipul de echipamente radio ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 este în conformitate cu Directiva 2014/53/UE.
Slovak	Aviat Networks týmto vyhlasuje, že rádiové zariadenie typu ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 je v súlade so smernicou 2014/53/EÚ.
Slovenian	Aviat Networks potrjuje, da je tip radijske opreme ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 skladen z Direktivo 2014/53/EU.
Spanish	Por la presente, Aviat Networks declara que el tipo de equipo radioeléctrico ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 es conforme con la Directiva 2014/53/UE.
Swedish	Härmed försäkrar Aviat Networks att denna typ av radioutrustning ODU 600/ODU 600sp with CTR 8311, CTR 8312, CTR 8380, CTR 8540 överensstämmer med direktiv 2014/53/EU.

Full declarations of conformity are available here:

- CTR 8311, 8312 with ODU 600v2: **https://view.highspot.com/viewer/60d35bfdf7794d471a56db18**
- CTR 8540 with ODU 600v2: **https://view.highspot.com/viewer/60d35c9a34d6be2c490aa999**

Country Availability Matrix

Aviat's radios are classified under the Radio Equipment Directive (2014/53/EU) as Class 2 products. For details of where the equipment is intended to be used, see the country matrix below. Aviat Networks intends to market this equipment where a cross (X) is shown.

Band (GHz)	**L6**	**U6**	**07**	**08**	**10**	**11**	**13**	**15**	**18**	**23**	**26**	**28**	**32**	**38**	**42**
Austria	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Belgium	X	X	X	X		X	X	X	X	X	X	X	X	X	
Bulgaria	X	X	X	X	X	X	X		X	X	X	X	X	X	
Croatia	X	X	X	X		X	X	X	X	X		X	X	X	X
Cyprus	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Czech Republic	X	X	X			X	X	X	X	X	X		X	X	
Denmark	X	X	X			X	X	X	X	X	X		X	X	X
Estonia	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Finland	X	X	X	X	X		X	X	X	X	X	X	X	X	X
France	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Germany	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Greece	X		X				X	X	X	X	X			X	
Hungary	X	X	X	X		X	X	X	X	X	X			X	
Iceland	X	X	X	X	X	X	X	X	X	X	X	X		X	
Ireland	X	X	X	X		X	X	X	X	X	X			X	
Italy	X	X	X	X	X	X	X	X	X	X	X		X	X	
Latvia	X	X	X	X			X	X	X	X	X	X		X	
Lithuania	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Luxembourg	X	X	X	X	X		X	X	X	X	X	X	X	X	
Malta	X	X	X	X	X	X	X	X	X	X			X	X	X
Netherlands	X	X	X				X	X	X	X	X	X	X	X	
Norway	X	X	X	X	X	X	X	X	X	X	X	X		X	X
Poland	X	X	X	X		X	X	X	X	X	X	X	X	X	X
Portugal	X	X	X	X		X	X	X	X	X	X			X	X
Romania	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Slovak Republic	X	X	X		X	X	X	X	X	X	X	X	X	X	
Slovenia	X	X	X	X		X	X	X	X	X	X	X	X	X	X
Spain	X	X		X	X	X	X	X	X	X	X	X		X	
Sweden	X	X	X	X			X	X	X	X	X		X	X	
Switzerland	X	X	X		X	X	X	X	X	X	X	X	X	X	X
United Kingdom	X	X	X			X	X	X	X	X	X	X	X	X	X

A license to operate this equipment will be required and the relevant regulator must be contacted prior to installation and commissioning.

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Chapter 1. IP Introduction

The Aviat Networks IP (Internet Protocol) is a protocol used for communicating data across a packet-switched internetwork using the Internet Protocol Suite, also referred to as TCP/IP.

IP is the primary protocol in the Internet Layer of the Internet Protocol Suite. It has the task of delivering distinguished protocol datagrams (packets) from the source host to the destination host solely based on their addresses.

This section describes the purpose and scope of the document, and conventions and acronyms used in this document.

Purpose and Scope

This document describes the basic and advanced configuration tasks of Aviat Networks Internet Protocol (IP). CLI (Command Line Interface) commands are used for the configuration. System administrators and users who configure and maintain Aviat Networks IP use this document.

This document describes the configuration of specific features, in a step by step manner using sample values. See the **CTR CLI Manuals** for a detailed explanation of the configuration parameters and the related values, if any.

Acronyms

Table 1: Acronyms Used in this Document

Acronym	Explanation
ARP	Address Resolution Protocol
BOOTP	BOOTstrap Protocol
CIDR	Classless Inter Domain Routing
DHCP	Dynamic Host Control Protocol
ICMP	Internet Control Message Protocol
IGMP	Internet Group Management Protocol
InARP	Inverse ARP
Id	Identifier
IP	Internet Protocol
IRDP	ICMP Router Discovery Protocol
MIB	Management Information Base
OSPF	Open Shortest Path First Protocol
PPP	Point-to-Point Protocol
RARP	Reverse Address Resolution Protocol
RFC	Request For Comments
SNMP	Simple Network Management Protocol
TCP	Transmission Control Protocol
TFTP	Trivial File Transfer Protocol
UDP	User Datagram Protocol
VAR	Value Added Reseller
VC	Virtual Circuit
MRP	Multicast Routing Protocol
RRD	Route Redistribution
RTM	Route Table Manager

References

- CTR 8500-8300 CLI User Manuals.
- See wikipedia for packet definition at http://en.wikipedia.org/wiki/Packet_%28information_technology%29

Chapter 2. IP General Configurations

The Aviat Networks IP is configurable and managed by CLI interfaces. This chapter describes the preliminary configurations of Aviat Networks IP.

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IP Default Configurations

Table 2: Default Configurations

Parameter	Default configuration
ICMP redirect messages	Enabled
ICMP unreachable messages	Enabled
ICMP echo replies	Enabled
ICMP mask replies	Enabled
IP rarp client request interval	100
IP rarp client request retries	4
Path MTU discovery	Disabled
Aggregate Routes	50
Multi paths	2
Load sharing	Disabled
Directed broadcast	Disabled
Proxy-ARP	Disabled
Proxy ARP subnet check	Enabled
Ping packet-size	40
Ping packet-count	3
Ping time-out	1 sec
IP vrf	Default
IP Routing	Enabled
IP Time–to–Live	64 sec
ARP Time out	300 sec
ARP Max retries	3
IPv4 enable	Enable
DHCP Relay Agent	Disabled

CLI configuration modes

This table lists the access and exit methods to the various CLI configuration modes.

Table 3: General Configurations

Command mode	Access method	Prompt	Exit method
User EXEC	This is the initial mode to start a session.	aos>	The logout method is used.
Privileged EXEC	Use the `enable` command to enter the Privileged EXEC mode.	aos#	Use the command **disable** to return from the Privileged EXEC mode to User EXEC mode, the disable command is used.
Global Configuration	Enter Privileged EXEC mode and use the command `configure terminal` to enter the Global Configuration Mode.	aos(config)#	Use the command **exit** to log out of the Global Configuration Mode, and the command **end** to log out of the Privileged EXEC mode.
Interface Configuration	In Global Configuration Mode, use the command `command interface <interface-type><interface-id>` to enter the Interface Configuration Mode.	aos(config-if)#	Use the command **exit** to log out of the Global Configuration Mode, and the command **end** to log out of the Privileged EXEC mode.

IP CLI command modes

Table 4: IP Specific CLI Command Modes

Command mode	Access method	Prompt	Exit method
Router Configuration	The protocol specific Global Configuration mode router command (such as `router rip`) is used to enter into the Router Configuration mode.	aos(config-router)#	The command `exit` is used to exit to the Global Configuration mode. The command `end` is used to exit to the Privileged EXEC mode.
VLAN (Virtual LAN) Configuration	The Global Configuration mode command `vlan <vlan #>` is used to enter the VLAN Configuration mode.	aos(config-vlan)#	Use the command `exit` to log out of the Global Configuration Mode, and the command `end` to log out of the Privileged EXEC mode.

Chapter 3. Configuring Aviat Networks IP

Aviat Networks IP supports dynamic registration of Higher Layer protocols, CIDR, subnets and static routes. It implements routing table using TRIE and maintains a common routing table using RTM to support route redistribution. This chapter describes the configuration of the following Aviat Networks IP features using CLI interfaces.

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Configuring the DHCP Relay Agent 23

Configuring IPv4 Address

You can assign IP addresses to network interfaces to enable those interfaces and allow communication with the hosts on those interfaces.

Note:

- An IP address can be a primary or secondary address.
- However, an interface can have only one primary IP address and multiple secondary addresses.
- Secondary IP addresses are not supported.

CLI configuration

Follow these steps to configure IPv4 address through CLI.

Step	Action

1	Enter the Interface Configuration mode.
	`aos#configure terminal`
	`aos(config)# interface vlan 1`

2	Configure IP address in vlan1 interface.
	`aos(config-if)# ip address 40.0.0.1 255.0.0.0`
	`aos(config-if)# no shutdown`

3	Execute the following show command to view the successful IP address configuration.
	`aos#show ip interface`
	`vlan1 is up, line protocol is up`
	`Internet Address is 40.0.0.1/8`
	`Broadcast Address  40.255.255.255`

4	To delete the IP address, use the no form of the command.
	`aos(config-if)#no ip address 40.0.0.1 255.0.0.0`
	`aos(config-if)#end`
	`aos#show ip interface`
	`vlan1 is up, line protocol is up`
	`Internet Address is 0.0.0.0/0`
	`Broadcast Address  255.255.255.255`

5	To delete the VLAN interface, use the no form of the following command. `aos(config)#no interface vlan 1`

Configuring IPv4 Routing

Enabling IPv4 routing allows you to configure Connection profiles (or similar profiles in an external authentication server) that define destinations across WAN interfaces and add routes to the routing table.

CLI configuration

Follow these steps to enable IP routing using CLI.

Step	Action
1	Enter the Global Configuration mode. `aos#configure terminal`
2	Enable IP routing. `aos(config)# ip routing`

3	Execute the following show command to view the IP routing information.
	`aos# show ip information`
	`VRF Name: default`
	`Global IP Configuration:`
	`----------`
	`IP routing is enabled`
	`Default TTL is 64`
	`ICMP redirects are always sent`
	`ICMP unreachables are always sent`
	`ICMP echo replies are always sent`
	`ICMP mask replies are always sent`
	`Number of aggregate routes is 50`
	`Number of multi-paths is 2`
	`Load sharing is disabled>`
	`Path MTU discovery is disabled`

4	To disable IP routing, use the no form of the command and view the output using the show command.
	`aos(config)# no ip routing`
	`aos# show ip information`
	`VRF Name: default`
	`Global IP Configuration:`
	`----------`
	`IP routing is disabled`
	`Default TTL is 64`
	`ICMP redirects are always sent`
	`ICMP unreachables are always sent`
	`ICMP echo replies are always sent`
	`ICMP mask replies are always sent`
	`Number of aggregate routes is 50`
	`Number of multi-paths is 2`
	`Load sharing is disabled`
	`Path MTU discovery is disabled`

Configuring Static IPv4 Routes

Static routing explicitly defines the next hop from a router to a particular destination. The Aviat Networks IP router provides a facility to define a static route to a destination. The destination is defined by a network prefix. This section describes how to configure Static IPv4 routes using CLI configuration.

CLI configuration

Follow these steps to add Static IPv4 routes using CLI.

In AOS1

Step	Action
1	Enter the Global Configuration mode. `aos#configure terminal`
2	Configure the route. `aos(config)# ip route 60.0.0.0  255.0.0.0 50.0.0.10`

3	Execute the following show command to view the route.
	`aos# show ip route`
	`Codes: C - connected, S - static, R - rip, B - bgp, O - ospf, I - IS-IS`
	`IA - OSPF inter area, N1 - OSPF NSSA external type 1,`
	`N2 - OSPF NSSA external type 2, E1 - OSPF external type 1,`
	`E2 - OSPF external type 2 L1 - IS-IS Level1, L2 - IS-IS Level2`
	`Vrf Name:  default`
	`---------`
	`C 40.0.0.0/8  is directly connected, vlan1`
	`C 50.0.0.0/8  is directly connected, vlan2`
	`S 60.0.0.0/8  [-1] via 50.0.0.10`

4	To delete a static route, use the no form of the command.
	`aos(config)# no ip route 60.0.0.0  255.0.0.0 50.0.0.10`
	`aos# show ip route`
	`Vrf Name:  default`
	`---------`
	`C 40.0.0.0/8  is directly connected, vlan1`
	`C 50.0.0.0/8  is directly connected, vlan2`

Configuring Static ARP Entries

Address Resolution Protocol (ARP) is used to connect Layer 3 IP address to Layer 2 MAC address.

It means that ARP is used to link our IP addressing to our Ethernet addressing (MAC Addressing).

If you want to communicate with any device on your network, you must have the Ethernet MAC address for that device. If the device is not on your LAN, then use the default gateway (your router). In this case, your router will be the destination MAC address that your PC will communicate with.

This configuration is used to add or delete a static ARP.

CLI configuration

The CLI configurations for static ARP entries performed through the Global Configuration mode as follows.

Step	Action
1	Enter the Global Configuration mode. `aos#configure terminal`
2	Configure the ARP cache timeout interval. `aos(config)# arp timeout 1000`
3	Configure the number of ARP request retry count. `aos(config)# ip arp max-retries 5`

Step	Action

4	View the ARP information.
	`aos# show ip arp information`
	`ARP Configurations:`
	`---------- ---------- ----------`
	`VRF Name: default`
	`Maximum number of ARP request retries is 5`
	`ARP cache timeout is 1000 seconds`

5	Configure the static ARP entry. `aos(config)# arp 40.123.31.78 00:11:22:33:47:0A vlan 1`

6	View the static ARP entries.
	`aos#show ip arp`
	`VRF Id : 0`
	`VRF Name: default`
	`Address Hardware Address Type Interface Mapping`
	`------- ---------- ---- --------- -------`
	`40.123.31.77 00:11:22:33:46:0a ARPA vlan1 Static`
	`40.123.31.78 00:11:22:33:47:0a ARPA vlan1 Static`

7	To delete a static ARP entry:
	`aos(config)# no arp 40.123.31.77`
	`aos#show ip arp`
	`VRF Id : 0`
	`VRF Name: default`
	`Address Hardware Address Type Interface Mapping`
	`------- ---------- ---- --------- -------`
	`40.123.31.78 00:11:22:33:47:0a ARPA vlan1 Static`

Sending IPv4 Ping

Ping is used to test whether a particular host is reachable across an IP network. Ping estimates the round-trip time and packet loss rate between hosts using interval timing and response rate.

The PING application is built over ICMP ECHO REQUEST and ECHO RESPONSE messages.

The results of the PING operations are stored in the PING table and the administrator can view it.

CLI configuration

Follow this step to send an IPv4 ping, using CLI.

Step	Action
1	Send ping to the destination (Host A of VLAN1 interface) from source (AOS1's VLAN interface) in the Global Configuration mode. `aos#ping 40.0.0.10` `Reply Received From :40.0.0.10, TimeTaken : 130 msecs` `Reply Received From :40.0.0.10, TimeTaken : 50 msecs` `Reply Received From :40.0.0.10, TimeTaken : 60 msecs` `--- 40.0.0.10 Ping Statistics ---` `3 Packets Transmitted, 3 Packets Received, 0% Packets Loss` `View the captured packets using Ethereal (the protocol analyzer).` `[root@localhost root]# tethereal -i eth1` `tethereal: Symbol `pcap_version' has different size in shared object, consider re-linking` `Capturing on eth1` `0.000000 3com_03:04:01 -> Broadcast ARP Who has 40.0.0.10? Tell 40.0.0.1` `0.015371 00:11:22:33:44:0a -> 3com_03:04:01 ARP 40.0.0.10 is at 00:11:22:33:44:0a` `0.038585 40.0.0.1 -> 40.0.0.10 ICMP Echo (ping) request` `0.065545 40.0.0.10 -> 40.0.0.1 ICMP Echo (ping) reply` `0.101199 40.0.0.1 -> 40.0.0.10 ICMP Echo (ping) request` `0.115553 40.0.0.10 -> 40.0.0.1 ICMP Echo (ping) reply` `0.135841 40.0.0.1 -> 40.0.0.10 ICMP Echo (ping) request` `0.155515 40.0.0.10 -> 40.0.0.1 ICMP Echo (ping) reply`

Configuring ICMP Messages

ICMP is an integral of any IP implementation. It is an error reporting and diagnostic utility used by routers, intermediary devices, or hosts to communicate updates or error information to other routers, intermediary devices, or hosts.

This section describes the configurations for enabling ICMP Redirect messages and Unreachable messages.

Enabling ICMP redirect message

ICMP redirect messages are sent to notify the hosts on the data link that a better route is available for a particular destination.

ICMP Redirect messages are generated when the destination can be reached through a gateway in the same network, and the packet was received by one other gateway in the same network.

The packet is forwarded to the particular gateway and not dropped.

CLI configuration

Follow these steps to enable ICMP redirect messages using CLI.

Step	Action
1	Enter the Global Configuration mode. `aos#configure terminal`
2	Enable the IP redirects. `aos(config)# ip redirects`

3	Execute the show command to view IP information.
	`aos#show ip information`
	`VRF  Name:  default`
	
	`Global IP Configuration:`
	`----------`
	`IP routing is enabled`
	`Default TTL is 64`
	`ICMP redirects are always sent`
	`ICMP unreachables are always sent`
	`ICMP echo replies are always sent`
	`ICMP mask replies are always sent`
	`Number of aggregate routes is 50`
	`Number of multi-paths is 2`
	`Load sharing is disabled`
	`Path MTU discovery is disabled`

4	To disable the ICMP redirects, execute the no form of the ip redirect command, use the show command to view the output.
	`aos(config)# no ip redirects`
	`aos#show ip information`
	`VRF  Name:  default`
	
	`Global IP Configuration:`
	`----------`
	`IP routing is enabled`
	`Default TTL is 64`
	`ICMP redirects are never sent`
	`ICMP unreachables are always sent`
	`ICMP echo replies are always sent`
	`ICMP mask replies are always sent`
	`Number of aggregate routes is 50`
	`Number of multi-paths is 2`
	`Load sharing is disabled`
	`Path MTU discovery is disabled`

Enabling ICMP unreachable message

ICMP Destination Unreachable messages are generated when any one of the following are not reachable:

- the destination network
- protocol
- application.

CLI configuration

Follow these steps to enable ICMP unreachable messages using CLI.

Step	Action
1	Enter the Global Configuration mode. `aos#configure terminal`
2	Enable the IP redirects. `aos(config)# ip unreachables`

3	Execute the show command to view IP information.
	`aos#show ip information`
	`VRF Name: default`
	
	`Global IP Configuration:`
	`----------`
	`IP routing is enabled`
	`Default TTL is 64`
	`ICMP redirects are always sent`
	`ICMP unreachables are always sent`
	`ICMP echo replies are always sent`
	`ICMP mask replies are always sent`
	`Number of aggregate routes is 50`
	`Number of multi-paths is 2`
	`Load sharing is disabled`
	`Path MTU discovery is disabled`

4	To disable the ICMP unreachable messages, execute the no form of the ip redirect command. Use the show command to view the output.
	`aos(config)# no ip unreachables`
	`aos#show ip information`
	`VRF Name: default`
	
	`Global IP Configuration:`
	`----------`
	`IP routing is enabled`
	`Default TTL is 64`
	`ICMP redirects are never sent`
	`ICMP unreachables are never sent`
	`ICMP echo replies are always sent`
	`ICMP mask replies are always sent`
	`Number of aggregate routes is 50`
	`Number of multi-paths is 2`
	`Load sharing is disabled`
	`Path MTU discovery is disabled`

Configuring IPv4 Trace Route

Trace route is an application built using UDP probes and ICMP (Time Exceeded and Port Unreachable) error messages to trace the route to the destination. It can help you identify:

- why connections to a given server are poor
- where exactly the problem is
- how systems are connected to each other.

Trace route works by increasing the time-to-live (TTL) value of each successive batch of packets sent. TTL is a limit on the period of time or number of iterations or transmissions that a unit of data (e.g. a packet) can experience before it should be discarded.

Configuring the DHCP Relay Agent

A DHCP Relay Agent is a host, or router, that forwards DHCP packets between clients and servers that are not located in the same physical subnet.

The DHCP relay agent will intercept DHCP packets received on, or addressed to, the configured IP interface's local subnet. It will then generate a new DHCP packet to forward it out of another interface.

DHCP-request packets will have the IP gateway address (GIADDR) field set to the IP address of the receiving interface before being forwarded to the configured DHCP server.

DHCP-reply packets are then broadcast back to the client via the subnet that the IP gateway address is a part of.

The DHCP relay agent on CTR can be configured to forward DHCP packets received on either a router port or an IP VLAN interface. Each interface can be configured to forward DHCP packets to up to 16 unique server addresses.

CLI configuration

Follow these steps on AOS1 to configure the DHCP Relay Agent to forward DHCP requests from the 40.0.0.0 subnet to the host at 60.0.0.10.

Step	Action
1	Enter the Global Configuration mode. `aos#configure terminal`
2	Enable the DHCP relay Agent. `aos(config)# ip dhcp relay-agent`
3	Enter the Interface Configuration mode. `aos(config)# interface vlan 1`
4	Configure the DHCP Relay Agent to forward all DHCP requests received on this interface to 60.0.0.10. `aos(config-if)# ip helper-address 60.0.0.10`

5	Execute the following show command to view the successful DHCP relay agent configuration.
	`aos# show ip dhcp relay`
	`Interface vlan 1:`
	`Helper Addresses: 60.0.0.10`

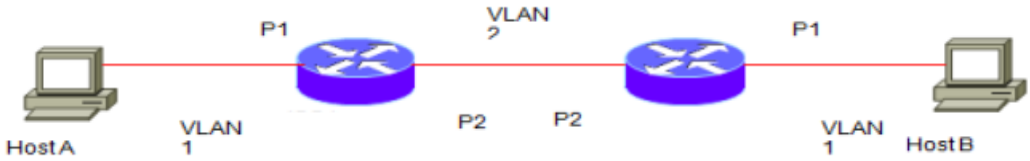
6	To stop the DHCP Relay Agent from forwarding DHCP packets to a server, use the no form of the command. `aos(config-if)# no ip helper-address 60.0.0.10`

Chapter 4. IP Topologies

This section provides sample deployment scenarios used for the configuration steps given in this document. The figure below depicts the sample topology used for configuring and testing the basic features of Aviat Networks IP. See the following topic for the command line configurations of these topologies.

Topology for Testing Aviat Networks IP

Figure 1: Topology for Aviat Networks IP



Note: P1 and P2 represent the port numbers of the CTR 8540 switches.

Table 5: IPv4 Addresses of Interfaces in the Switches and Hosts – Aviat Networks IP

Switch / Host	Interface of CTR 8540 Switches	IPv4 Address / Mask
AOS1 (AOS switch)	VLAN 1	40.0.0.1 / 255.0.0.0
	VLAN 3	50.0.0.1 / 255.0.0.0
AOS2 (AOS switch)	VLAN 3	50.0.0.10 / 255.0.0.0
	VLAN 1	60.0.0.1 / 255.0.0.0
Host A (host device connected AOS1)		40.0.0.1 / 255.0.0.0
Host B (host device connected AOS2)		60.0.0.1 / 255.0.0.0

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